Zoe allgems: unified architecture and verification

Zoe / allgems

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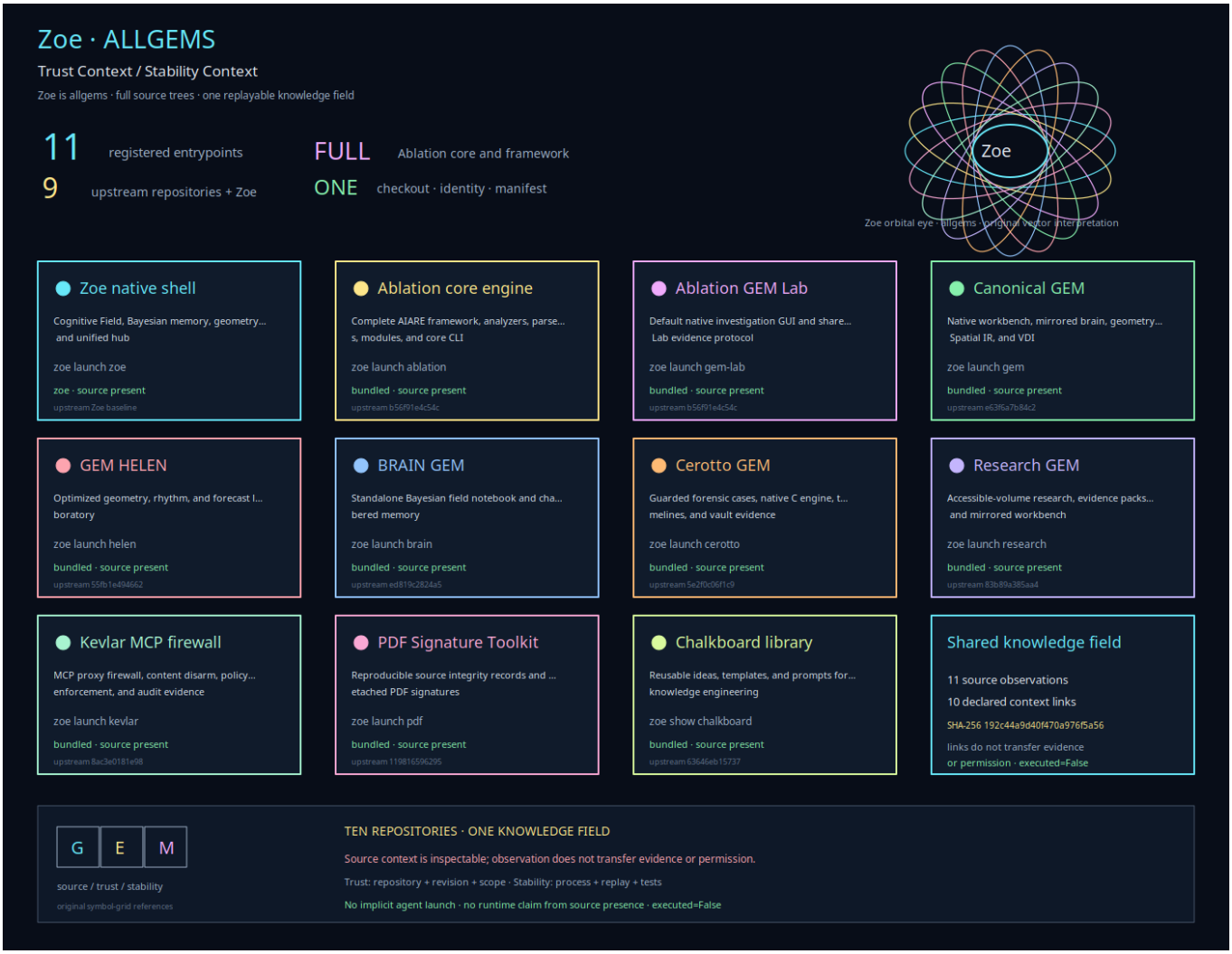


Figure 1: Zoe allgems: unified architecture and verification · visual study

Zoe is a single source repository containing the ten public Ablation-Tool organization repositories, including

Zoe. The original repositories remain independent upstream projects. This repository does not require submodules or a runtime download to inspect their source.

Embedded layouts can resolve a host component through `ZOE_ABLATION_ROOT`, `ZOE_CEROTTO_ROOT`, or `ZOE_CHALKBOARD_ROOT`. Bundled sources take precedence. Host sources are used only when their component anchors exist. Chalkboard remains a content library, not a GUI process.

0.1 Source and provenance

`components.lock.json` records the upstream repository and full revision for every imported source tree. The snapshots were imported with `git subtree --squash`, preserving their content, licenses, notices, and source-revision metadata.

Source	Revision	Included role
Zoe	`f30c981` baseline	Native Cognitive Field, Bayesian memory, geometry, and unified hub
Ablation	`b56f91e4c54cfd996bceea63a38d813dc1d089fc`	Complete core engine, framework, native GEM Lab, tests, docs, and assets
graphics-environment-manager	`e63f6a7b84c2e8fffc0fcf01001986aeaaaffcb5`	Complete native GEM and mathematical workbench
GEM HELEN	`55fb1e494662e9b69e964ac2f9df4172dfffed0c`	Complete native geometry and forecast experiments
BRAIN GEM	`ed819c2824a55b019a532f4fa2cb44853d3d1b27`	Complete Bayesian field notebook
Cerotto	`5e2f0c06f1c9b6f6fe7d2489175ce89e7ab6ed4`	Native forensic engine and guarded GEM workbench
next-token-predictability	`83b89a385aa4e67fb0c1b60ba30a21ec0e5b0f9`	Accessible volume research and saved evidence
Kevlar	`8ac3e0181e98572d6c41f964385393a4441af2a`	MCP server and pure policy engine
PDF Signature Toolkit	`1198165962958c5132130f3854626ff30c333941`	Source integrity and PDF signature commands
Chalkboard	`63646eb15737d822316073ba1f2114d0857cc72`	Lessons, templates, and prompts; content only

Eleven launch or inspection endpoints span those ten repositories because Ablation supplies both its core CLI and native Lab. Chalkboard has an inspection endpoint, not a fictitious GUI command.

0.2 The mesh

`guiai.gem.knowledge.KnowledgeField` is the shared event memory. It records:

- Source observations with component, repository, revision, source, and explicit scope.
- Stability context identifying component-specific processes, unverified runtime state, and canonical event replay.
- Declared links between observations and components.
- `belief_updated=False`, `executed=False`, `evidence_transfer=False`, and `permission_transfer=False` where applicable.

A source observation records **source registration only**. It does not prove runtime behavior, truth, trustworthiness, or permission to act. Declared links are inspectable relationships, not evidence or authorization transfer. The shared graph uses canonical JSON, sequence numbers, previous-event hashes, SHA-256, complete load-time verification, and no-overwrite saves. It provides tamper evidence, not authentication.

Trust Context means repository, revision, source, scope, and the relevant authorization boundary. **Stability Context** means isolated component processes, pinned source revisions, deterministic replay, and separately tested behavior. Neither phrase promises automatic trust, availability, correctness, or security.

0.3 Native hub and CLI

Install the hub command from this checkout:

```
python -m pip install ./guiai-lab
zoe
```

The native selector is **Zoe · allgems**. It supports:

- Opening Zoe's Cognitive Field.
- Launching the Ablation core CLI, native Ablation Lab, canonical GEM, HELEN, BRAIN, Cerotto, research workbench, Kevlar, and PDF verification.
- Inspecting Chalkboard's content source without starting a process.
- Recording source-context observations and declared links.
- Exporting a component manifest or the knowledge graph.

The default native Ablation Lab command is `gem launch --no-agent`. Cerotto launches only after a case database, case ID, and conversation ID are supplied. No component starts merely because the hub or its scene was opened.

The selector also opens built-in creative modules: **Zoe Artline**, **Zoe Pentomino**, **Zoe Practice Navigator**, **Zoe Resilience Atlas**, **Zoe Presence**, **Zoe Protocol Lab**, **Zoe Waveform Replay**, and **Zoe Patchbay**. These are part of the Zoe source tree, not additional upstream repositories or new component endpoints. They retain the ten-repository manifest and eleven existing source endpoints. Their editors, clocks, exports, and replay actions run only after explicit user interaction. Waveform Replay reads bounded VCD, legacy JSONL, normalized JSON, and optional WDB evidence only after an explicit path or command is supplied; matching signatures are not proof of signal identity or hardware behavior. Patchbay inspects firmware only after an explicit path, signs new update capsules through a separate CLI action, and requires both an execution flag and a hardware-write gate for applicable acceptance plans. Its native Cog produces review-only token forecasts, not commands. The root Declaration and Constitution define the project's agency, honesty, privacy, accessibility, provenance, and review commitments.

Equivalent command examples:

```
zoe list
zoe show chalkboard
zoe launch ablation -- --help
zoe launch gem-lab
zoe launch gem
zoe launch helen
zoe launch brain
zoe launch research
zoe launch kevlar -- --help
zoe launch pdf -- --help
zoe launch cerotto -- --db case.db --case incident-7 --conversation conv-7
```

```
zoe manifest --output gem-components.json
zoe knowledge --output gem-knowledge.json
```

Arguments after `--` are forwarded as an argument array without a shell. Component working directories and Python paths are explicit. The host's unrelated `PYTHONPATH` is not inherited. `ZOE_ROOT`, `ZOE_PYTHON`, and `ZOE_PYTHON_DEPENDENCIES` can select the checkout, interpreter, and additional dependency directories. Each component retains its own dependency manifest; the hub does not install every optional model, parser, or native tool.

0.4 Verification

Unified integration and real headless command checks:

```
GEM_COMPONENT_TESTS=1 PYTHONPATH=guiai-lab \
  /usr/bin/python3 -m unittest stress.unified_stress -v

PYTHONPATH=guiai-lab guiai-lab/stress/nested_x11.sh \
  /usr/bin/python3 -m unittest stress.unified_stress -v
```

The suite checks all ten source trees, full Ablation engine presence, manifest revisions, trust/stability metadata, graph replay, edited and reordered event detection, no-overwrite saves, content-library inspection, command isolation, actual component CLI calls, native launch and source-link controls, and Spatial IR/VDI verification.

Component-specific verification uses each project's own commands. Cerotto's normal and sanitizer suites are required. Kevlar uses its full `pytest` suite. Canonical GEM and Zoe use their core and native X11 suites. Research uses the frozen evidence verifiers and the canonical 17-operation workbench check. PDF Signature Toolkit is exercised with a temporary synthetic LaTeX source record, not a real identity or production signature. Chalkboard is checked as content, including its documented template paths.

Measured results for this integration:

- Ablation core: **2803 passed, 93 skipped, 49 subtests passed**.
- Ablation Lab: **69 core, 36 GTK, 1 real desktop-input, 30 native Tk, and 35 GEM tests passed**. The 40-tab churn test completed with bounded memory and all child processes reaped.
- Canonical GEM: **65 integration tests passed**, plus **30 tool-discovery checks in both core and native runs** and the **17-operation workbench check**.
- Cerotto: **65 normal and 65 sanitizer tests passed**. Kevlar: **149 tests passed**. BRAIN: **16 tests passed**, including native workflows.
- Research: **16 comparison cases** and **120 BioBERT cases verified**. PDF seal/verify and no-overwrite checks used a synthetic source record. Chalkboard template and prompt paths were checked.
- Unified integration: **8 headless and 8 native tests passed**, including real component commands, source graph replay, native controls, and VDI checks.
- Zoe regressions: **31 existing tests with four optional skips**, **28 cognitive/GEM native regressions**, and **8 headless plus 8 native unified tests passed**.
- Package: both wheel and source distribution built; the wheel installed into the isolated test runtime. The installed `zoe` entrypoint opened the native hub used for the screenshot below. The exported graph reloaded with 11 observations, 10 links, and 21 events. The actual VDI file passed `verify_vdi` in `vdi-verification.json`.

Product-facing text uses `Zoe` without changing lowercase commands, imports, or environment identifiers. Embedded host components resolve through their documented root overrides, including `ZOE_CHALKBOARD_ROOT`. BRAIN uses the existing sibling `_callbacks` helper correctly. Ablation Lab reuses each button's themed

pixbuf, updates its image only when that pixbuf changes, restores compact preferences synchronously, and lets the shutdown fallback reap an exited child even when the VTE callback has not run. No original test assertion was weakened.

`evidence/unified/verification.json` records these results and their scopes. `components.json` and `source-graph.json` contain the final source identity. `render-*.json`, `vdi-verification.json`, and `chalkboard.json` preserve the corresponding real exports in that directory. One unrelated process-stat race appeared once in `test_timeout_kills_the_entire_process_group`: `/proc/<pid>` disappeared between the test's existence check and read. The unchanged suite passed on rerun; its assertion was not modified.

No single command is presented as proof that every optional backend, model, graphics driver, external VDI consumer, or production security boundary has been validated. Source presence and source registration do not imply runtime verification, trustworthiness, or authorization.

0.5 Update discipline

Update one component at a time with its upstream URL and a reviewed revision, for example:

```
git subtree pull --prefix=components/ablation \
  https://github.com/Ablation-Tool/ablation.git main --squash
```

Do not bulk-update all components without reviewing their changes. Update `components.lock.json`, preserve local work, rerun the affected component and unified suites, and record new evidence. Never force-push over the upstream projects.

0.6 Emblem and symbol references

`assets/branding/zoe-eye.png` is the primary Zoe emblem, generated from the user-supplied eye/orbital reference by `assets/branding/render_eye.py`. The earlier `orbital-emblem.png` remains the source of the orbital motif. The G, E, and M symbol-grid references are preserved under `assets/branding/references/` with their source filenames in `assets/branding/PROVENANCE.md`. Historical recognition for Gary Kildall and the original Digital Research contributors is recorded in `NOTICE`.

The screenshot shows the actual GTK hub and explicit component controls. Source presence is not runtime verification. Component processes, evidence, and authorization remain distinct even though their source and trust/stability context meet in one repository and knowledge graph.

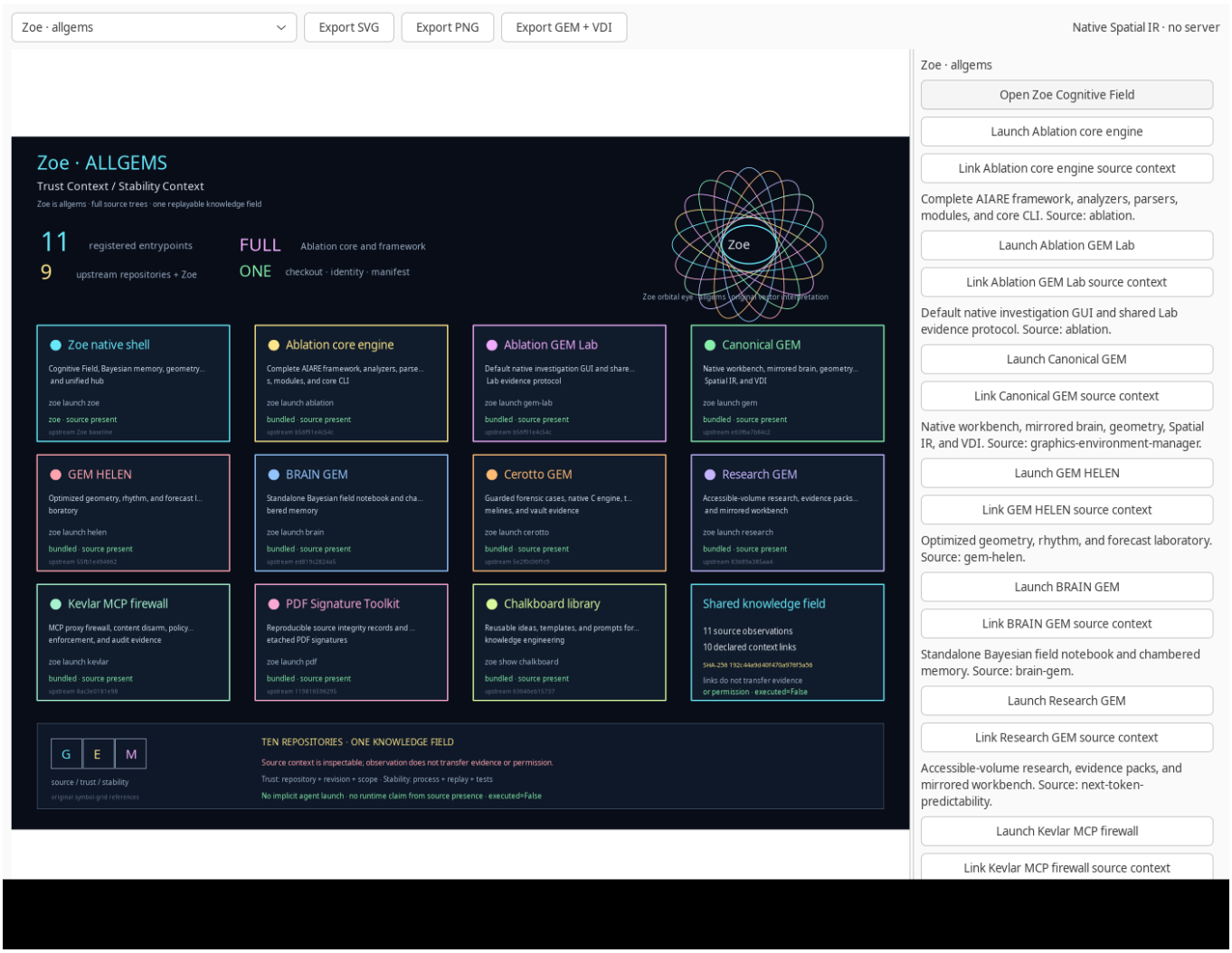


Figure 2: Native unified hub